How selection changes ensemble RPF

A matched comparison on 12,104 models across three residue masks. Native RPF evaluates 50-conformer panels with the current TR peak lists.

TM80 S1 had higher DP in all 30 matched comparisons, with increases of +0.09 to +0.21. Changing the clustering prior gave -0.01 to +0.10 in the original screen. The three-seed follow-up found an average MCMC50 minus Gumbel50 DP of +0.009.

Native ensemble DP by context, seed 1

Residues / pool / clustering prior	Uniform 50	Gumbel 50	Exact 50	MCMC50	DISCO50
1-139 / All / Primary	0.61	0.60	0.57	0.56	0.56
1-139 / All / HDBSCAN	0.61	0.62	0.63	0.66	0.60
1-139 / TM80 S1 / Primary	0.72	0.74	0.74	0.74	0.75
1-139 / TM80 S1 / HDBSCAN	0.72	0.74	0.73	0.76	0.76
5-95 / All / Primary	0.61	0.60	0.57	0.63	0.56
5-95 / All / HDBSCAN	0.61	0.62	0.63	0.66	0.62
5-95 / TM80 S1 / Primary	0.73	0.75	0.75	0.74	0.74
5-95 / TM80 S1 / HDBSCAN	0.73	0.75	0.74	0.75	0.74
32-47 + 78-99 / All / Primary	0.61	0.60	0.57	0.55	0.56
32-47 + 78-99 / All / HDBSCAN	0.61	0.63	0.64	0.63	0.59
32-47 + 78-99 / TM80 S1 / Primary	0.76	0.76	0.76	0.76	0.76
32-47 + 78-99 / TM80 S1 / HDBSCAN	0.76	0.77	0.76	0.76	0.76

Each cell shows one prespecified seed. Scores are printed to 0.01. Primary is component-validated UMAP KNN. The clustering prior only affects selectors that use it.

Matched differences versus Uniform 50

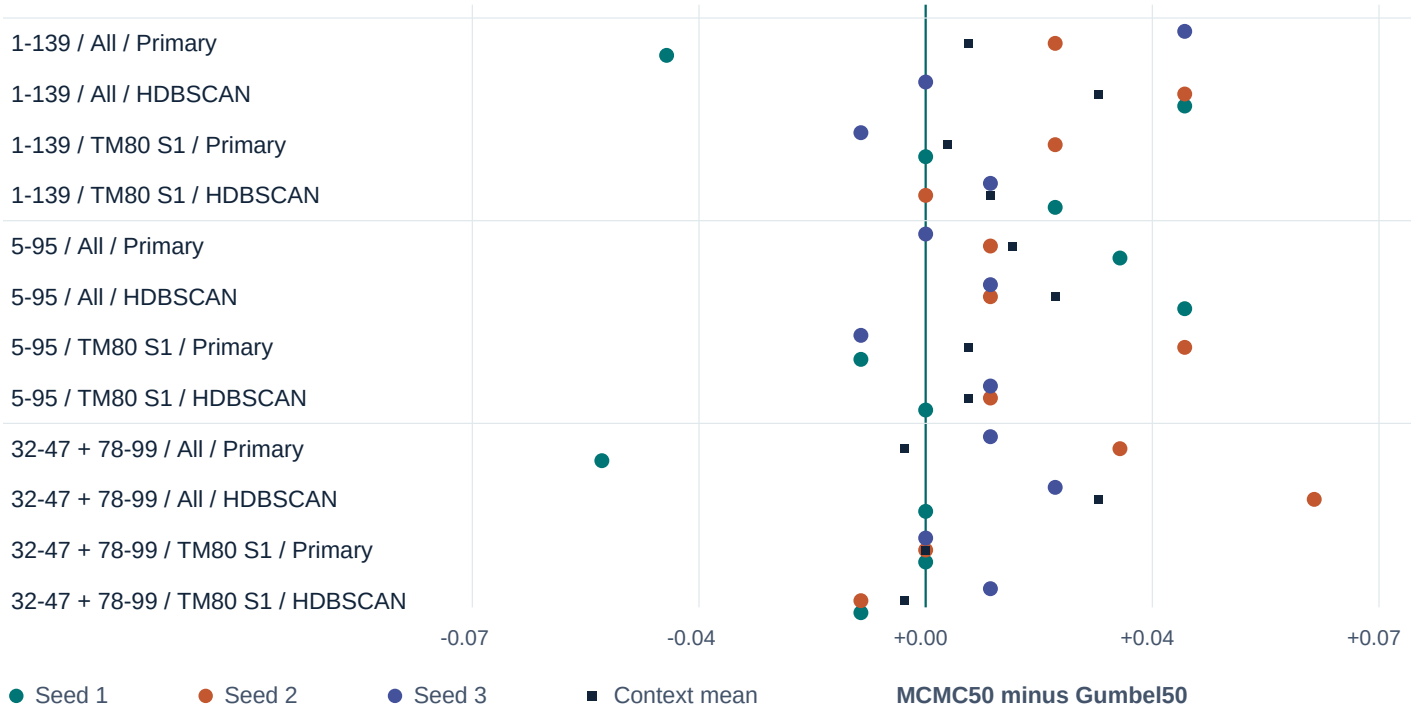
Selector	Mean DP delta	Range of DP deltas	Higher / tie / lower
Gumbel 50	+0.008	-0.01 to +0.02	8 / 1 / 3
Exact 50	+0.001	-0.04 to +0.03	7 / 2 / 3
MCMC50	+0.010	-0.06 to +0.05	8 / 2 / 2
DISCO50	-0.007	-0.05 to +0.04	5 / 2 / 5

MCMC50 versus Gumbel50 across seeds

36 matched seed-context pairs. Each pair uses the same residue mask, clustering prior, eligible pool and peak lists.

Mean DP was **0.679** for MCMC50 and **0.669** for Gumbel50. The paired mean favored MCMC50 (+0.009 for MCMC50 minus Gumbel50). MCMC50 was higher in 20 pairs, Gumbel50 in 7, and 9 tied. Individual differences ranged from -0.05 to +0.06.

Paired native DP difference by context



Seed	MCMC50 mean	Gumbel50 mean	Mean DP gap	Range of paired gaps
1 (202611710)	0.683	0.682	+0.002	-0.05 to +0.04
2 (202611711)	0.671	0.652	+0.019	-0.01 to +0.06
3 (202611712)	0.682	0.675	+0.007	-0.01 to +0.04

Mean within-context seed SD was 0.016 for MCMC50 and 0.020 for Gumbel50. Mean paired recall and precision differences were +0.003 and +0.002. These are descriptive contrasts from three seeds on overlapping model pools. Native scores are printed to 0.01.

Scoring rewards different selections

Whole-cohort results pooled over the six mask-by-prior contexts. Stochastic rows average three seeds per context. Deterministic controls contribute one selection per context.

Standalone scores of selected models

Selector	Models	RPF DP	Recall	Precision	MolProbity	ProteinEBM
Uniform 50	50	0.654	0.936	0.887	3.10	4,083.2
Gumbel 50	50	0.644	0.933	0.885	2.86	3,978.9
Exact 50	50	0.659	0.934	0.890	2.75	3,960.0
MCMC50	50	0.655	0.933	0.889	2.76	3,960.9
DISCO50	50	0.649	0.930	0.889	2.88	3,952.9
Energy minimum / cluster	33-152	0.642	0.927	0.889	3.04	3,993.9
Top RPF DP	50	0.791	0.959	0.927	1.68	3,889.7
Lowest MolProbity	50	0.761	0.954	0.918	1.19	3,905.5
Lowest ProteinEBM	50	0.722	0.955	0.899	1.69	3,703.1

Higher RPF is favorable. Lower MolProbity and lower ProteinEBM t=0.10 are favorable within this cohort. Direct ranking controls are evaluated on the same scores used to select them. Historical energy minimum / cluster keeps its variable cardinality.

Occupancy estimates count repeat visits

Sampler	Distinct*	RPF DP	Recall	Precision	MolProbity	ProteinEBM
Uniform visits	6,324	0.675	0.939	0.893	2.91	4,051.0
Exact categorical visits	4,927	0.657	0.934	0.889	2.79	3,963.7
Metropolis visits	3,713	0.659	0.934	0.890	2.78	3,965.3

*Mean distinct models among 9,000 draws or retained chain visits per context and seed. These rows are visit-weighted summaries. Distinct-model ensembles are reported separately.

Weight sensitivity

The historical Gumbel grid used cluster-size exponents 0, 0.5 and 1 and energy scales of 0.5, 1 and 2 cohort IQRs. Across its 162 panels, mean member DP spanned 0.490-0.724 and mean energy spanned 3,847.4-4,120.4. The ranges include mask, prior and seed variation.

The state rule changes eligibility

Fixed-anchor TM threshold sensitivity uses the same bidirectional native score columns. State labels are local to each residue mask.

Residues	TM60 eligible	TM70 eligible	TM80 eligible	Largest Primary	Largest HDBSCAN
1-139	11,722 (97%)	11,665 (96%)	11,492 (95%)	10,854	123
5-95	11,720 (97%)	11,637 (96%)	11,353 (94%)	9,188	154
32-47 + 78-99	8,578 (71%)	5,574 (46%)	2,097 (17%)	11,043	159

TM eligibility sums disjoint states with at least 50 members. TM cores are 15-72 for the first two masks and 32-47 for the discontinuous mask. Largest-cluster controls use the clustering mask and exclude HDBSCAN noise. TM60 and TM70 retain the audited TM80 anchors, so they measure threshold sensitivity.

What ran from Git history

ff465df174 / 21459bf220 Uniform, Gumbel weighted panels, exact categorical and Metropolis occupancy, and minimum-energy model per assigned cluster.

975260f126 Exact product-weight subset and fixed-size swap MCMC. Both use the same target law.

53145b5c16 Residual TM state membership and DISCO50, using cluster representatives plus a weighted fill.

The original CSP_Rank trace producer is missing from the recovered history. Original best-pTM DISCO requires an audited pTM vector covering the same cohort. The energy-based representative methods above are the recovered implementations with complete inputs. Archived Ward analyses use different model registries.

Comparison design and score provenance

1,824 selection evaluations used audited standalone RPF, MolProbity and ProteinEBM for all eligible pools. The native panel comparison crossed both clustering priors with all models and TM80 S1 for each mask. Native panel inputs used sorted model IDs. Native ASDP 2.3 Model ave uses midrange inter-proton distances. Standalone RPF summaries use per-model scores. The same original-control peak lists (SR215-TR-2026-11-06-v1) and verified protonated coordinates were used throughout. Audits 24234 and 24326 bind clustering and native TM. Audit 24390 verified 52 distinct native panel calls. The follow-up audit 24430 reused the first seed and scored the two other frozen seeds.